Answer **all** the questions in the spaces provided.

1 Identification of an unknown organic acid by titration

In this experiment, you will perform titrations to determine the relative molecular mass, M_r, of an unknown monoprotic organic acid, **FA 1,** and use it to identify the acid.

FA 1 is an aqueous solution containing 4.80 g dm^{-3} of an unknown monoprotic organic acid.

You are also provided with:
FA 2 is 0.110 mol dm^{-3} sodium hydroxide, NaOH.
thymol blue indicator

Prepare a table in the space provided below to record, to an appropriate level of precision, all your burette readings.

(a) Titration of FA 1 with FA 2

1. Fill the burette with **FA 2**.
2. Pipette 25.0 cm^3 of **FA 1** into a conical flask.
3. Add 3 to 4 drops of thymol blue indicator into the flask. The solution in the flask should turn pink.
4. Titrate **FA 1** in the conical flask with **FA 2**. The end-point is reached when the solution changes from yellow to green.
5. Record your burette readings in your table.
6. Repeat steps 2 to 5 until consistent results are obtained.

Titration Results

[5]

From your titrations, obtain a suitable volume of **FA 2**, to be used in your calculations. Show clearly how you obtained this volume.

volume of **FA 2** = ……………………….. [1]

(b) **(i)** Calculate the amount of sodium hydroxide present in the volume of **FA 2** that you calculated in **(a)**.

amount of NaOH reacted = ……………………. [1]

(ii) Calculate the concentration of organic acid present, in mol dm^{-3}, in **FA 1**.

.

concentration of organic acid in **FA 1** = ……………………….. [1]

(iii) Calculate the relative molecular mass, M_r, of the organic acid in **FA 1**.

M_r of the organic acid = ……………………….. [1]

(iv) From another experiment, the identity of **FA 1** was narrowed down to these four acids.

CH_3COOH $HCOOH$ $CH_2{=}CHCO_2H$ $CH_3CH_2CO_2H$

Given that sodium hydroxide reacts only with the COOH group in the organic acid, deduce the identity of the organic acid present in **FA 1**. Explain your answer.

[A_r: H, 1.0; C, 12.0; O, 16.0; C*l*, 35.5]

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………………………………………………………………………………………………… [1]

(c) This identification method uses the calculated relative molecular mass of the acid.

As the relative molecular masses of $CH_2{=}CHCO_2H$ and $CH_3CH_2CO_2H$ are so similar, any slight inaccuracy in the titration could lead to an incorrect conclusion.

Describe a chemical test that would enable you to distinguish between $CH_2{=}CHCO_2H$ and $CH_3CH_2CO_2H$.

There is **no need** to carry out this test.

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(d) Another student was provided a solution of another unknown organic acid. It has the same concentration, in mol dm^{-3}, as that used in the titration above. However, this new unknown organic acid is a diprotic acid.

State and explain how the calculated relative molecular mass obtained will differ from that obtained in **b(iii)**.

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[Total: 14]

2 Determination of the concentration of NaOH and the enthalpy change of neutralisation, ΔH_{neut}

You are provided with the following solutions:

FA 3 is a solution of sodium hydroxide, NaOH

FA 4 is a solution of 2.00 mol dm^{-3} hydrochloric acid, HC*l*

In this question, you are to perform a series of 6 experiments where different volumes of **FA 3** and **FA 4** are mixed together to give a total volume of 50 cm^3. The temperature change, ΔT, for each experiment will be determined and a graph of ΔT against the volume of **FA 3** will be plotted.

You will then use the data from your graph to determine the concentration of NaOH in **FA 3**, and the enthalpy change of neutralisation for the reaction between aqueous sodium hydroxide and hydrochloric acid.

$$NaOH + HCl \rightarrow NaCl + H_2O \qquad \Delta H_{neut}$$

(a) Procedure

Experiment 1

1) Place one polystyrene cup inside a second polystyrene cup. Place these into a 250 cm^3 beaker to prevent them from tipping over.
2) Use a 50 cm^3 measuring cylinder to transfer 10.0 cm^3 of **FA 3** into the polystyrene cup.
3) Measure the temperature of **FA 3** in the polystyrene cup and record the initial temperature of **FA 3**, T_1, in Table 2.1 on page 7.
4) Use another 50 cm^3 measuring cylinder, transfer 40.0 cm^3 of **FA 4** into the same polystyrene cup.
5) Stir the mixture in the polystyrene cup with the thermometer. Measure and record the highest temperature, T_2 in Table 2.1 on page 7.
6) Rinse the polystyrene cup and thermometer with distilled water and dry them with paper towels.
7) Repeat steps **2** to **6** using 20.0 cm^3, 30.0 cm^3 and 40.0 cm^3 of **FA 3**, each time using appropriate volume of **FA 4**, such that the total volume of the reaction mixture is 50 cm^3.
8) Record all measurements of volume, temperature (T_1 and T_2) and temperature change, ΔT, in Table 2.1 on page 7.

(b) Results

Table 2.1

	experiment					
	1	**2**	**3**	**4**	**5**	**6**
volume of **FA 3** / cm^3	10.0	20.0	30.0	40.0		
volume of **FA 4** / cm^3	40.0					
initial temperature, T_1/ ^oC						
final temperature, T_2/ ^oC						
temperature change, ΔT/ ^oC						

[3]

(c) (i) Plot a graph of ΔT (y–axis) against volume of **FA 3** used (x–axis) using the four experimental results that you have obtained.

The scales for both axes must be chosen to provide an origin.

By considering your plotted points, perform two additional experiments to identify the volume of **FA 3** needed to produce the maximum temperature change, ΔT_{max}.

In each experiment, ensure that the total volume of the reaction mixture is 50 cm^3. You may find it helpful to plot the results from each experiment before choosing the volumes to use in the next experiment.

Record all measurements of volume, temperature (T_1 and T_2) and temperature change, ΔT, in Table 2.1 on page 7. [2]

(ii) Draw two straight lines of best fit. The first best fit line should be drawn using the plotted points where ΔT is increasing and the second best fit line should be drawn using the plotted points where ΔT is decreasing. Extrapolate these two lines until they cross. [1]

(d) **(i)** Use your graph to determine the maximum temperature change, ΔT_{max}, as well as the volume of **FA 3,** V_{FA3}**,** used to obtain it.

ΔT_{max} = ………………………………. °C

V_{FA3} = ..……...………………… cm^3 [1]

(ii) Using your results in **(d)(i)**, calculate the concentration of NaOH in **FA 3**.

concentration of NaOH in **FA 3** = ……………………………… [2]

(iii) Hence, calculate the enthalpy change of neutralisation, ΔH_{neut}.

You may assume that the specific heat capacity of the reaction mixture is 4.18 J g^{-1} K^{-1}, and that the density of the reaction mixture is 1.00 g cm^{-3}.

ΔH_{neut} = ……………………………….. [2]

(e) Predict and explain the following:

(i) the effect on ΔT_{max} when the volumes of **FA 3** and **FA 4** used in the reaction are doubled.

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(ii) the effect on ΔH_{neut} if the experiment was repeated with ethanoic acid, CH_3CO_2H, of the same concentration instead of hydrochloric acid.

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(f) State one significant source of error in the experiment and suggest an improvement that can be made to reduce this error.

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[Total: 14]

3 Inorganic Analysis

FA 5 is a solution which contains up to two cations and one anion.

Carry out the following tests and carefully record your observations in Table 3.1.

Using the observations in Table 3.1, you will then identify the ions present in **FA 5**.

Unless otherwise stated, the volumes given below are approximate and should be estimated rather than measured. Test and identify any gases evolved.

Table 3.1

		test	observations
(a)	**(i)**	Place one drop of **FA 5** solution on Universal Indicator paper.	
	(ii)	Add about 1 cm^3 of dilute nitric acid to 1 cm depth of **FA 5** solution. Add barium nitrate dropwise until no further change is seen.	
	(iii)	Add sodium hydroxide to 1 cm depth of **FA 5** solution until no further change is seen. Gently warm the mixture.	

(iv)	Add aqueous ammonia to 1 cm depth of **FA 5** until no further change is seen.	
(v)	Add I^-(aq) to 1 cm depth of **FA 5** solution until no further change is seen. Leave the mixture to stand.	

[5]

(b) **(i)** Identify the cation(s) and anion present in **FA 5**.

cation(s) in **FA 5** ……………………………………………………….……….........

anion in **FA 5…**……………………………………………………………………. [3]

(ii) Write an equation to explain the observation in test **(a)(i)**.

……………………………………………………………………………………. [1]

[Total: 9]

4 Determination of oxidising strength of oxidising agents

You are provided with samples of the following four aqueous solutions.

solution that contains bromide ions, Br^-
solution that contains iron(III) ions, Fe^{3+}
solution that contains aqueous iodine, I_2
solution that contains iodide ions, I^-

Carry out the tests in Table 4.1 to investigate possible redox reactions and rank the oxidising strength of the oxidising agents Br_2, Cl_2, Fe^{3+} and I_2.

$$Br_2 + 2e^- \rightleftharpoons 2Br^-$$
$$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$$
$$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$$
$$I_2 + 2e^- \rightleftharpoons 2I^-$$

Your answers should correspond to the given observations/equations/deductions. The results of the first test have been given to you and you should use it as an example.

Table 4.1

		test	observations	deductions
(a)	**(i)**	Transfer 1 cm^3 of Br^- to a test-tube. Add 1 cm^3 of Cl_2.	Solution turns yellow.	Cl_2 oxidises Br^- to Br_2 $Cl_2 > Br_2$
	(ii)	Transfer 1 cm^3 of Fe^{3+} to a test-tube. Add 1 cm^3 of Br^-. Add aqueous NaOH dropwise until in excess.		
	(iii)	Transfer 1 cm^3 of Fe^{3+} to a test-tube. Add 1 cm^3 of I^- then add 5 drops of starch solution.		
	(iv)	Transfer 1 cm^3 of I_2 to a test-tube. Add 1 cm^3 of Br^- then add 5 drops of starch solution.		

[4]

(b) Use your information in Table 4.1 to rank the order of oxidising strengths of the species: Br_2, Cl_2, Fe^{3+}, I_2.

Note: In some of the tests performed in part **(a)** there will have been no reaction. Such tests can still help you to deduce the relative oxidising powers of the species involved.

.... [1]

(c) Using your observations, explain why aqueous FeI_3 cannot be prepared.

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.... [1]

[Total: 6]

5 Planning

When a solute is added to two immiscible solvents, **A** and **B**, some of the solute dissolves in each of the solvents and an equilibrium is set up between the two solvents. It has been shown that for dilute solutions, at equilibrium, the ratio of the two concentrations is a constant known as the partition coefficient, $K_{partition}$.

Trichloromethane, $CHCl_3$, and water separate into two immiscible layers when shaken together and allowed to stand. Ammonia can dissolve in both of these layers. The $K_{partition}$ for ammonia in these two solvents is given by:

$$K_{partition} = \frac{[NH_3]_{water}}{[NH_3]_{trichloromethane}}$$

50 cm^3 of trichloromethane and 50 cm^3 of aqueous ammonia are mixed and left to equilibrate for about an hour. Samples of the aqueous layer are obtained and the amount of ammonia present in the aqueous layer is determined by titration. The concentration of ammonia in the separate layers can be calculated and the value of $K_{partition}$ can then be determined.

(a) Explain why ammonia is likely to be more soluble in water than in trichloromethane.

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……[1]

(b) You are to plan a procedure that will allow you to obtain the titration results necessary to determine value of $K_{partition}$ of ammonia between water and trichloromethane at room temperature.

You are provided with the following
- trichloromethane, $CHCl_3$,
- 1.00 mol dm^{-3} aqueous ammonia,
- 0.500 mol dm^{-3} hydrochloric acid,
- methyl orange indicator,
- 250 cm^3 conical flask with stopper,
- 10.0 cm^3 pipette,
- the laboratory apparatus normally found in a school.

In your plan you should include the following:

- an outline of how you would prepare the equilibrium mixture,
- practical details of how you would carry out the titration,
- appropriate apparatus and their capacities.

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(c) It is unnecessary to titrate both layers of the partition. Explain why the titration is carried out on a sample of the aqueous layer rather than the trichloromethane layer.

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(d) A student carried out the experiment and obtained a mean titre of 19.20 cm^3.

(i) Using this mean titre value, calculate the equilibrium amounts of ammonia

1. in the aqueous layer
2. in the trichloromethane layer

amount of ammonia in the aqueous layer = ………………………….[1]

amount of ammonia in the trichloromethane layer = ………………………….[1]

(ii) Hence calculate the value of the partition coefficient, $K_{partition}$.

partition coefficient, $K_{partition}$ = ………………………………………….[1]

[Total: 12]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	$NaOH(aq)$	$NH_3(aq)$
aluminium, $Al^{3+}(aq)$	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, $NH_4^+(aq)$	ammonia produced on heating	–
barium, $Ba^{2+}(aq)$	no ppt. (if reagents are pure)	no ppt.
calcium, $Ca^{2+}(aq)$	white. ppt. with high [$Ca^{2+}(aq)$]	no ppt.
chromium(III), $Cr^{3+}(aq)$	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), $Cu^{2+}(aq)$	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), $Fe^{2+}(aq)$	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), $Fe^{3+}(aq)$	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, $Mg^{2+}(aq)$	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), $Mn^{2+}(aq)$	off-white ppt. rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, $Zn^{2+}(aq)$	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

anion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^- (aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^- (aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^- (aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^- (aq)	NH_3 liberated on heating with OH^-(aq) and Al foil
nitrite, NO_2^- (aq)	NH_3 liberated on heating with OH^-(aq) and Al foil; NO liberated by dilute acids (colourless NO → (pale) brown NO_2 in air)
sulfate, SO_4^{2-} (aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-} (aq)	SO_2 liberated on warming with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple